THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE CHAPTER TEST SET A

Chapter 4: **Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A change in which only physical properties such as shape, size or state change is a: 1

(a) nuclear change **(b)** combustion **(c)** chemical change **(d)** physical change

2. Rusting of iron is a: 1

(a) chemical change **(b)** evaporation
(c) reversible change **(d)** physical change

3. A reaction in which a substance reacts with oxygen giving heat and light is: 1

(a) combustion **(b)** sublimation **(c)** neutralisation **(d)** evaporation

4. Fireflies glow because of: 1

(a) electric current **(b)** heating
(c) a chemical change in their body **(d)** reflection of sunlight

5. Which is a reversible change? 1

(a) Rusting **(b)** Cooking rice
(c) Burning of coal **(d)** Freezing of water

6. The process of formation of crystals from a solution is: 1

(a) combustion **(b)** crystallisation **(c)** neutralisation **(d)** rusting

7. Which of these is a slow natural change? 1

(a) Bursting of firecrackers **(b)** Lightning
(c) Burning of paper **(d)** Weathering of rocks

8. Changes which are good for us, e.g. ripening fruits, are: 1

(a) harmful **(b)** undesirable **(c)** desirable **(d)** slow only

9. A candle melting is ___, while the burning wick is ___. 1

(a) physical, physical **(b)** chemical, physical
(c) physical, chemical **(d)** chemical, chemical

10. A fire extinguisher works by: 1

(a) cutting off oxygen supply **(b)** adding fuel
(c) raising temperature **(d)** adding sugar

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. Distinguish between a physical and a chemical change with an example. 2

12. What is combustion? Give two combustible substances. 2

13. What is ignition temperature? 2

14. What is bioluminescence? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an activity to show that burning of magnesium is a chemical change. 3

16. Explain the conditions necessary for combustion. 3

17. Classify the changes: melting of butter, burning of candle, cooking egg, dissolving salt. 3

OR

Describe the experiment of passing carbon dioxide through lime water. 3

18. Why is rusting harmful? How can it be prevented? 3

OR

What are the differences between physical and chemical changes (three points)? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Define physical and chemical changes.
(ii) Give three examples of each.
(iii) How do you identify a chemical change? 5

OR

Case study: A kitchen shows milk boiling, curd setting, ice melting and iron tawa rusting. 5
(i) Classify.
(ii) Which can be reversed?
(iii) Why does iron rust?
(iv) How can it be prevented?

20. (i) What is combustion?
(ii) Describe the three conditions.
(iii) Describe three ways to extinguish fire. 5

OR

Case study: A candle burns on a table. A beaker is placed over it and the flame goes out after a while. 5
(i) Why?
(ii) Which condition is removed?
(iii) Which change occurred?
(iv) How is it used in fire safety?

END OF QUESTION PAPER — Total: 40 Marks

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